JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Height and Distance

JEE Main Mathematics · Previous Year Questions with Answer Key

42	41	2004 – 2023	41	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer)	1 question
Out of syllabus	41 questions
Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 1 questions

Q1. JEE Main 2020 (Online) 4th September Morning Slot

Two vertical poles AB = 15 m and CD = 10 m are standing apart on a horizontal ground with points A and C on the ground. If P is the point of intersection of BC and AD, then the height of P (in m) above the line AC is :

(A) 10/3 (B) 5 (C) 20/3 (D) 6

Out of syllabus 41 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q2. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

The angle of elevation of the top P of a tower from the feet of one person standing due South of the tower is 45° and from the feet of another person standing due west of the tower is 30°. If the height of the tower is 5 meters, then the distance (in meters) between the two persons is equal to

(A) 10 (B) $\frac{5}{2}\sqrt{5}$ (C) $5\sqrt{5}$ (D) 5

Q3. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

From the top A of a vertical wall AB of height $30\ \mathrm{m}$, the angles of depression of the top P and bottom Q of a vertical tower PQ are 15° and 60° respectively, B and Q are on the same horizontal level. If C is a point on AB such that $\mathrm{CB}=\mathrm{PQ}$, then the area (in m^2) of the quadrilateral BCPQ is equal to :

(A) $200(3-\sqrt{3})$ (B) $300(\sqrt{3}-1)$ (C) $300(\sqrt{3}+1)$ (D) $600(\sqrt{3}-1)$

Q4. JEE Main 2022 (Online) 29th July Morning Shift · MCQ (Single Correct Answer)

The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is $\cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$. If the distance of the point B from the tower is 15 units, then $\cot\alpha$ is equal to :

(A) $\frac{6}{5}$ (B) $\frac{9}{5}$ (C) $\frac{4}{3}$ (D) $\frac{7}{3}$

Q5. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

A horizontal park is in the shape of a triangle OAB with $\mathrm{AB}=16$. A vertical lamp post OP is erected at the point O such that $\angle\mathrm{PAO}=\angle\mathrm{PBO}=15^{\circ}$ and $\angle\mathrm{PCO}=45^{\circ}$, where C is the midpoint of AB. Then $(\mathrm{OP})^2$ is equal to :

(A) $\frac{32}{\sqrt{3}}(\sqrt{3}-1)$ (B) $\frac{32}{\sqrt{3}}(2-\sqrt{3})$ (C) $\frac{16}{\sqrt{3}}(\sqrt{3}-1)$ (D) $\frac{16}{\sqrt{3}}(2-\sqrt{3})$

Q6. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45°. Let R be a point on AQ and from a point B, vertically above R, the angle of elevation of P is 60°. If $\angle BAQ = 30^\circ$, $AB = d$ and the area of the trapezium PQRB is α, then the ordered pair (d, α) is :

(A) $(10(\sqrt{3}-1), 25)$ (B) $\left(10(\sqrt{3}-1), \frac{25}{2}\right)$

(C) $(10(\sqrt{3}+1), 25)$ (D) $\left(10(\sqrt{3}+1), \frac{25}{2}\right)$

Q7. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Let a vertical tower AB of height $2h$ stand on a horizontal ground. Let from a point P on the ground a man can see upto height h of the tower with an angle of elevation 2α. When from P, he moves a distance d in the direction of $\overrightarrow{AP}$, he can see the top B of the tower with an angle of elevation α. If $d = \sqrt{7}h$, then $\tan\alpha$ is equal to

(A) $\sqrt{5}-2$ (B) $\sqrt{3}-1$ (C) $\sqrt{7}-2$ (D) $\sqrt{7}-\sqrt{3}$

Q8. JEE Main 2022 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

A tower PQ stands on a horizontal ground with base Q on the ground. The point R divides the tower in two parts such that $QR = 15$ m. If from a point A on the ground the angle of elevation of R is 60° and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is :

(A) $5(2\sqrt{3}+3)$ m (B) $5(\sqrt{3}+3)$ m (C) $10(\sqrt{3}+1)$ m (D) $10(2\sqrt{3}+1)$ m

Q9. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

From the base of a pole of height 20 meter, the angle of elevation of the top of a tower is 60°. The pole subtends an angle 30° at the top of the tower. Then the height of the tower is :

(A) $15\sqrt{3}$ (B) $20\sqrt{3}$ (C) 20 + $10\sqrt{3}$ (D) 30

Q10. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

Let AB and PQ be two vertical poles, 160 m apart from each other. Let C be the middle point of B and Q, which are feet of these two poles. Let $\frac{\pi}{8}$ and θ be the angles of elevation from C to P and A, respectively. If the height of pole PQ is twice the height of pole AB, then $\tan^2\theta$ is equal to

(A) $\frac{3-2\sqrt{2}}{2}$ (B) $\frac{3+\sqrt{2}}{2}$ (C) $\frac{3-2\sqrt{2}}{4}$ (D) $\frac{3-\sqrt{2}}{4}$

Q11. JEE Main 2021 (Online) 31st August Morning Shift · MCQ (Single Correct Answer)

A vertical pole fixed to the horizontal ground is divided in the ratio 3 : 7 by a mark on it with lower part shorter than the upper part. If the two parts subtend equal angles at a point on the ground 18 m away from the base of the pole, then the height of the pole (in meters) is :

(A) $12\sqrt{15}$ (B) $12\sqrt{10}$ (C) $8\sqrt{10}$ (D) $6\sqrt{10}$

Q12. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

Two poles, AB of length a metres and CD of length a + b ($b \neq a$) metres are erected at the same horizontal level with bases at B and D. If BD = x and $\tan\angle ACB = \frac{1}{2}$, then :

(A) $x^2 + 2(a + 2b)x - b(a + b) = 0$

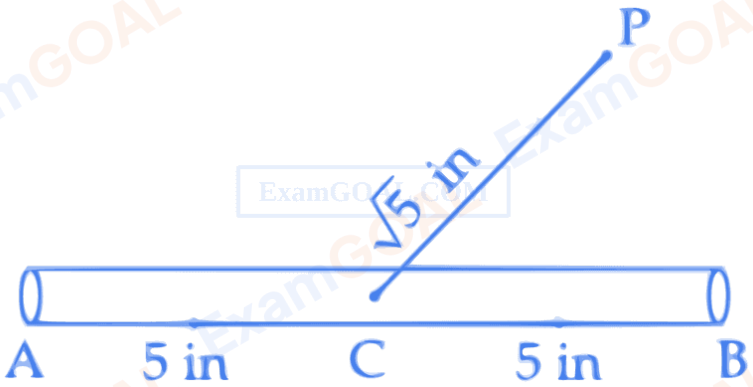
(B) $x^2 + 2(a + 2b)x + a(a + b) = 0$

(C) $x^2 - 2ax + b(a + b) = 0$

(D) $x^2 - 2ax + a(a + b) = 0$

Q13. JEE Main 2021 (Online) 26th August Evening Shift · MCQ (Single Correct Answer)

A 10 inches long pencil AB with mid point C and a small eraser P are placed on the horizontal top of a table such that PC = $\sqrt{5}$ inches and $\angle PCB = \tan^{-1}(2)$. The acute angle through which the pencil must be rotated about C so that the perpendicular distance between eraser and pencil becomes exactly 1 inch is :



(A) $\tan^{-1}\left(\frac{3}{4}\right)$

(B) $\tan^{-1}(1)$

(C) $\tan^{-1}\left(\frac{4}{3}\right)$

(D) $\tan^{-1}\left(\frac{1}{2}\right)$

Q14. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

A spherical gas balloon of radius 16 meter subtends an angle 60° at the eye of the observer A while the angle of elevation of its center from the eye of A is 75°. Then the height (in meter) of the top most point of the balloon from the level of the observer's eye is :

(A) $8(2 + 2\sqrt{3} + \sqrt{2})$

(B) $8(\sqrt{6} + \sqrt{2} + 2)$

(C) $8(\sqrt{2} + 2 + \sqrt{3})$

(D) $8(\sqrt{6} - \sqrt{2} + 2)$

Q15. JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

Let in a right angled triangle, the smallest angle be θ. If a triangle formed by taking the reciprocal of its sides is also a right angled triangle, then $\sin\theta$ is equal to :

(A) $\frac{\sqrt{5}+1}{4}$

(B) $\frac{\sqrt{5}-1}{2}$

(C) $\frac{\sqrt{2}-1}{2}$

(D) $\frac{\sqrt{5}-1}{4}$

Q16. JEE Main 2021 (Online) 18th March Evening Shift · MCQ (Single Correct Answer)

A pole stands vertically inside a triangular park ABC. Let the angle of elevation of the top of the pole from each corner of the park be $\frac{\pi}{3}$. If the radius of the circumcircle of ΔABC is 2, then the height of the pole is equal to :

(A) $\frac{1}{\sqrt{3}}$ (B) $2\sqrt{3}$ (C) $\sqrt{3}$ (D) $\frac{2\sqrt{3}}{3}$

Q17. JEE Main 2021 (Online) 25th February Morning Shift · MCQ (Single Correct Answer)

A man is observing, from the top of a tower, a boat speeding towards the lower from a certain point A, with uniform speed. At that point, angle of depression of the boat with the man's eye is 30° (Ignore man's height). After sailing for 20 seconds, towards the base of the tower (which is at the level of water), the boat has reached a point B, where the angle of depression is 45°. Then the time taken (in seconds) by the boat from B to reach the base of the tower is :

(A) $10(\sqrt{3}+1)$ (B) $10(\sqrt{3}-1)$ (C) 10 (D) $10\sqrt{3}$

Q18. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

The angle of elevation of a jet plane from a point A on the ground is 60°. After a flight of 20 seconds at the speed of 432 km/hour, the angle of elevation changes to 30°. If the jet plane is flying at a constant height, then its height is :

(A) $3600\sqrt{3}$ m (B) $1200\sqrt{3}$ m (C) $1800\sqrt{3}$ m (D) $2400\sqrt{3}$ m

Q19. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

Two vertical poles are 150 m apart and the height of one is three times that of the other. If from the middle point of the line joining their feet, an observer finds the angles of elevation of their tops to be complementary, then the height of the shorter pole (in meters) is :

(A) 30 (B) 25 (C) $20\sqrt{3}$ (D) $25\sqrt{3}$

Q20. JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

The angle of elevation of the summit of a mountain from a point on the ground is 45°. After climbing up one km towards the summit at an inclination of 30° from the ground, the angle of elevation of the summit is found to be 60°. Then the height (in km) of the summit from the ground is :

(A) $\frac{1}{\sqrt{3}-1}$ (B) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

(C) $\frac{1}{\sqrt{3}+1}$ (D) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

Q21. JEE Main 2020 (Online) 4th September Evening Slot · MCQ (Single Correct Answer)

The angle of elevation of a cloud C from a point P, 200 m above a still lake is 30°. If the angle of depression of the image of C in the lake from the point P is 60°,then PC (in m) is equal to :

(A) $200\sqrt{3}$ (B) 400 (C) 100 (D) $400\sqrt{3}$

Q22. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

The angle of elevation of the top of a vertical tower standing on a horizontal plane is observed to be 45° from a point A on the plane. Let B be the point 30 m vertically above the point A. If the angle of elevation of the top of the tower from B be 30°, then the distance (in m) of the foot of the tower from the point A is :

(A) $15\left(1+\sqrt{3}\right)$ (B) $15\left(3-\sqrt{3}\right)$ (C) $15\left(3+\sqrt{3}\right)$ (D) $15\left(5-\sqrt{3}\right)$

Q23. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

ABC is a triangular park with AB = AC = 100 metres. A vertical tower is situated at the mid-point of BC. If the angles of elevation of the top of the tower at A and B are $\cot^{-1}(3\sqrt{2})$ and $\text{cosec}^{-1}(2\sqrt{2})$ respectively, then the height of the tower (in metres) is :

(A) $\frac{100}{3\sqrt{3}}$ (B) 25 (C) 20 (D) $10\sqrt{5}$

Q24. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

Two poles standing on a horizontal ground are of heights 5m and 10 m respectively. The line joining their tops makes an angle of 15º with ground. Then the distance (in m) between the poles, is :-

(A) $5\left(2+\sqrt{3}\right)$ (B) $\frac{5}{2}\left(2+\sqrt{3}\right)$

(C) $10\left(\sqrt{3}-1\right)$ (D) $5\left(\sqrt{3}+1\right)$

Q25. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

Two vertical poles of heights, 20 m and 80 m stand a part on a horizontal plane. The height (in meters) of the point of intersection of the lines joining the top of each pole to the foot of the other, from this horizontal plane is :

(A) 12 (B) 16 (C) 15 (D) 18

Q26. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

If the angle of elevation of a cloud from a point P which is 25 m above a lake be 30° and the angle of depression of reflection of the cloud in the lake from P be 60°, then the height of the cloud (in meters) from the surface of the lake is :

(A) 45 (B) 42 (C) 50 (D) 60

Q27. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

Consider a triangular plot ABC with sides AB = 7m, BC = 5m and CA = 6m. A vertical lamp-post at the mid point D of AC subtends an angle 30° at B. The height (in m) of the lamp-post is -

(A) $2\sqrt{21}$ (B) $\frac{3}{2}\sqrt{21}$ (C) $7\sqrt{3}$ (D) $\frac{2}{3}\sqrt{21}$

Q28. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

A man on the top of a vertical tower observes a car moving at a uniform speed towards the tower on a horizontal road. If it takes 18 min. for the angle of depression of the car to change from 30° to 45° ; then after this, the time taken (in min.) by the car to reach the foot of the tower, is :

(A) $9\left(1+\sqrt{3}\right)$ (B) $18\left(1+\sqrt{3}\right)$

(C) $18\left(\sqrt{3}-1\right)$ (D) $\frac{9}{2}\left(\sqrt{3}-1\right)$

Q29. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

PQR is a triangular park with PQ = PR = 200 m. A T.V. tower stands at the mid-point of QR. If the angles of elevation of the top of the tower at P, Q and R are respectively 45°, 30° and 30°, then the height of the tower (in m) is :

(A) $50\sqrt{2}$ (B) 100 (C) 50 (D) $100\sqrt{3}$

Q30. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

A tower T_1 of height 60 m is located exactly opposite to a tower T_2 of height 80 m on a straight road. Fromthe top of T_1, if the angle of depression of the foot of T_2 is twice the angle of elevation of the top of T_2, then the width (in m) of the road between the feetof the towers T_1 and T_2 is :

(A) $10\sqrt{2}$ (B) $10\sqrt{3}$ (C) $20\sqrt{3}$ (D) $20\sqrt{2}$

Q31. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

An aeroplane flying at a constant speed, parallel to the horizontal ground, $\sqrt{3}$ kmabove it, is obsered at an elevation of 60° from a point on the ground. If, after five seconds, its elevation from the same point, is 30°, then the speed (in km / hr) of the aeroplane, is :

(A) 1500 (B) 1440 (C) 750 (D) 720

Q32. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

Let a vertical tower AB have its end A on the level ground. Let C be the mid-point of AB and P be a point on the ground such that AP = 2AB. If $\angle$BPC = β, then tanβ is equal to:

(A) $\frac{1}{4}$ (B) $\frac{2}{9}$ (C) $\frac{4}{9}$ (D) $\frac{6}{7}$

Q33. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The angle of elevation of the top of a vertical tower from a point A, due east of it is 45°. The angle of elevation of the top of the same tower from a point B, due south of A is 30°. If the distance between A and B is $54\sqrt{2}\,m$, then the height of the tower (in metres), is :

(A) $36\sqrt{3}$ (B) 54 (C) $54\sqrt{3}$ (D) 108

Q34. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30°. After walking for 10 minutes from A in the same direction, at a point B, he observes that the angle of elevation of the top of the pillar is 60°. Then the time taken (in minutes) by him, from B to reach the pillar, is :

(A) 6 (B) 10 (C) 20 (D) 5

Q35. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

If the angles of elevation of the top of a tower from three collinear points A, B and C, on a line leading to the foot of the tower, are 30°, 45° and 60° respectively, then the ratio, $AB : BC$, is :

(A) $1 : \sqrt{3}$ (B) $2 : 3$ (C) $\sqrt{3} : 1$ (D) $\sqrt{3} : \sqrt{2}$

Q36. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

A bird is sitting on the top of a vertical pole 20 m high and its elevation from a point O on the ground is 45°. It flies off horizontally straight away from the point O. After one second, the elevation of the bird from O is reduced to 30°. Then the speed (in m/s) of the bird is :

(A) $20\sqrt{2}$ (B) $20\left(\sqrt{3}-1\right)$

(C) $40\left(\sqrt{2}-1\right)$ (D) $40\left(\sqrt{3}-\sqrt{2}\right)$

Q37. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

$ABCD$ is a trapezium such that AB and CD are parallel and $BC \perp CD$. If $\angle ADB = \theta$, $BC = p$ and $CD = q$, then AB is equal to:

(A) $\frac{(p^2+q^2)\sin\theta}{p\cos\theta+q\sin\theta}$

(B) $\frac{p^2+q^2\cos\theta}{p\cos\theta+q\sin\theta}$

(C) $\frac{p^2+q^2}{p^2\cos\theta+q^2\sin\theta}$

(D) $\frac{(p^2+q^2)\sin\theta}{(p\cos\theta+q\sin\theta)^2}$

Q38. AIEEE 2008 · MCQ (Single Correct Answer)

AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle of elevation of the point A from a certain point C on the ground is 60°. He moves away from the pole along the line BC to a point D such that $CD = 7$ m. From D the angle of elevation of the point A is 45°. Then the height of the pole is :

(A) $\frac{7\sqrt{3}}{2}\frac{1}{\sqrt{3}-1}m$ (B) $\frac{7\sqrt{3}}{2}\left(\sqrt{3}+1\right)m$

(C) $\frac{7\sqrt{3}}{2}\left(\sqrt{3}-1\right)m$ (D) $\frac{7\sqrt{3}}{2}\frac{1}{\sqrt{3}+1}m$

Q39. AIEEE 2007 · MCQ (Single Correct Answer)

A tower stands at the centre of a circular park. A and B are two points on the boundary of the park such that $AB(=a)$ subtends an angle of 60° at the foot of the tower, and the angle of elevation of the top of the tower from A or B is 30°. The height of the tower is :

(A) $a/\sqrt{3}$ (B) $a\sqrt{3}$ (C) $2a/\sqrt{3}$ (D) $2a\sqrt{3}$

Q40. AIEEE 2004 · MCQ (Single Correct Answer)

A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank of the river is 60° and when he retires 40 meters away from the tree the angle of elevation becomes 30°. The breadth of the river is :

(A) $60\ m$ (B) $30\ m$ (C) $40\ m$ (D) $20\ m$

Q41. JEE Main 2020 (Online) 6th September Morning Slot · Numerical

Let AD and BC be two vertical poles
at A and B respectively on a horizontal ground.
If AD = 8 m, BC = 11 m and AB = 10 m; then the distance
(in meters) of a point M on AB from the point A such
that $MD^2 + MC^2$ is minimum is _____.

Q42. JEE Main 2020 (Online) 6th September Morning Slot · Numerical

The angle of elevation of the top of a hill from a point on the horizontal plane passing through the foot of the hill is found to be 45°. After walking a distance of 80 meters towards the top, up a slope inclined at an angle of 30° to the horizontal plane, the angle of elevation of the top of the hill becomes 75°. Then the height of the hill (in meters) is _____.

Answer Key

MCQ (Single Correct Answer)

1. **D**				

Out of syllabus

2. **A**	3. **D**	4. **A**	5. **B**	6. **A**
7. **C**	8. **A**	9. **D**	10. **C**	11. **B**
12. **C**	13. **A**	14. **B**	15. **B**	16. **B**
17. **A**	18. **B**	19. **D**	20. **A**	21. **B**
22. **C**	23. **C**	24. **A**	25. **B**	26. **C**
27. **D**	28. **A**	29. **B**	30. **C**	31. **B**
32. **B**	33. **B**	34. **D**	35. **C**	36. **B**
37. **A**	38. **B**	39. **A**	40. **D**	41. **5**
42. **80**				